Example 6.6 If θ is a parameter, find the equation of the locus of a moving point, whose coordinates are $(a(\theta - \sin\theta)\,,\ \ a(1-\cos\theta))$.

Solution:

Let $P\ (h,\ k)$ be any point on the required path. From the given information we have

$$h = a(\theta - \sin\theta) \qquad (6.1)$$
$$k = a(1-\cos\theta) \qquad (6.2)$$

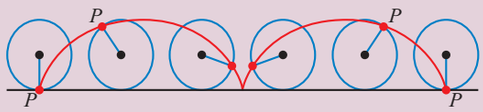


Figure 6.12

Let us find the value of θ and $\sin\theta$ from equation (6.2)

$$k = a(1-\cos\theta)$$

$$\cos\theta = \frac{a-k}{a} \ \Rightarrow\ \theta = \cos^{-1}\left(\frac{a-k}{a}\right) \text{ and } \sin\theta = \frac{\sqrt{2ak-k^2}}{a}$$

Substituting above values in (6.1) we get $h = a\cos^{-1}\left(\frac{a-k}{a}\right) - \sqrt{2ak-k^2}$

The locus of $(h,\ k)$ is

$$x = a\cos^{-1}\left(\frac{a-y}{a}\right) - \sqrt{2ay-y^2} \qquad (6.3)$$

https://www.geogebra.org/b/bd2ADu2I#material/zCKMj8kE

Though, the parametric form given above is converted to Cartesian form, in some cases the parametric form may be more useful to work with than the cartesian form.

Exercise - 6.1

1. Find the locus of P, if for all values of α, the co-ordinates of a moving point P is
   $(i)\ (9\cos\alpha\,,\ 9\sin\alpha)\ \ (ii)\ \ (9\cos\alpha\,,\ 6\sin\alpha)$.
2. Find the locus of a point P that moves at a constant distant of (i) two units from the x-axis (ii) three units from the y-axis.
3. If θ is a parameter, find the equation of the locus of a moving point, whose coordinates are $x = a\cos^3\theta$, $y = a\sin^3\theta$.
4. Find the value of k and b, if the points $P(-3,1)$ and $Q(2,b)$ lie on the locus of $x^2 - 5x + ky = 0$.
5. A straight rod of length 8 units slides with its ends A and B always on the x and y axes respectively. Find the locus of the mid point of the line segmentAB
6. Find the equation of the locus of a point such that the sum of the squares of the distance from the points $(3,5)$, $(1,-1)$ is equal to 20
7. Find the equation of the locus of the point P such that the line segment AB, joining the points $A(1,-6)$ and $B(4,-2)$, subtends a right angle at P.
8. If O is origin and R is a variable point on $y^2 = 4x$, then find the equation of the locus of the mid-point of the line segment OR.
9. The coordinates of a moving point P are $\left(\frac{a}{2}(\text{cosec}\theta + \sin\theta)\ ,\ \frac{b}{2}(\text{cosec}\theta - \sin\theta)\right)$, where θ is a variable parameter. Show that the equation of the locus P is $b^2x^2 - a^2y^2 = a^2b^2$.
10. If $P(2,-7)$ is a given point and Q is a point on $2x^2 + 9y^2 = 18$, then find the equations of the locus of the mid-point of PQ.